BUILDING AIR LEAKAGE DIAGNOSTIC TEST
BUILDING ENCLOSURES AND DWELLING UNIT ENCLOSURES

CALIFORNIA ENERGY COMMISSION

CEC-CF2R-ENV-20-H

SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

CERTIFICATE OF INSTALLATION

Note: This table completed by ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. Enclosure Air Leakage – General Information

01	Test Procedure used	
02	Date of the Diagnostic Test for this Dwelling	
03	Is ECC verification of building enclosure air leakage to outside required by CF1R?	
04	Target Enclosure Air Leakage from CF1R (CFM50)	
05	Indoor temperature during test (degrees Fahrenheit (°F))	
06	Outdoor temperature during test (degrees Fahrenheit (°F))	
07	Blower Door Location	
08	Building Elevation Above Sea Level (feet (ft))	

B. Diagnostic Equipment Information

01	Number of Manometers Used to Measure Home Pressurization	

02	03	04	05	06
Manometer Make	Manometer Model	Manometer Serial Number	Manometer Calibration Date	Manometer Calibration Status

07	Number of Fans Used to Pressurize Home	

08	09	10	11
Fan Make	Fan Model	Fan Serial Number	Fan Configuration (rings)

C1. Enclosure Air Leakage Diagnostic Test for a Single-Point Test with Manual Meter

01	Time Average Period of Meter (seconds)	
02	Test Methodology	
03	Pre-Test Baseline Enclosure Pressure (Pa) (May be positive or negative)	
04	Unadjusted Enclosure Pressure Target (Pa)	
05	Unadjusted Enclosure Pressure Measured (Pa) (Pressurization is positive; Depressurization is negative)	
06	Induced Enclosure Pressure Difference (Pa) Goal = 50 ± 3 or -50 ± 3 (Pressurization is positive; Depressurization is negative)	
07	Induced Enclosure Pressure Check	
08	Measured Nominal Fan Flow at Above Fan Pressure (cfm) at the Induced Enclosure Pressure Difference (in C06 above)	
09	Calculated Nominal CFM50	

C2. Enclosure Air Leakage Diagnostic Test for a Single-Point Test with Automatic Meter

01	Time Average Period of Meter (seconds)	
02	Test Methodology	
03	Pre-Test Baseline Enclosure Pressure (Pa) (May be positive or negative)	
04	Induced Enclosure Pressure from Manometer (Pa) Goal = 50 ± 3 or -50 ± 3 (Pressurization is positive; Depressurization is negative)	
05	Induced Enclosure Pressure Check	
06	Nominal CFM50	

C3. Enclosure Air Leakage Diagnostic Test for a Multi-Point Test

01	Time Average Period of Meter (seconds)	
02	Test Methodology	
03	Pre-Test Baseline Enclosure Pressure (Pa) (May be positive or negative)	
04	Unadjusted Enclosure Pressure Target (Pa)	
05	Unadjusted Enclosure Pressure Measured (Pa) (Pressurization is positive; Depressurization is negative)	
06	Induced Enclosure Pressure Difference (Pa) Goal = 60 ± 3 or -60 ± 3 (Pressurization is positive; Depressurization is negative)	
07	A minimum of five readings were taken spaced evenly between 10 Pa and 60 Pa (or highest attainable pressure)	
08	Post-Test Baseline Enclosure Pressure (Pa)	
09	Name and Version of ASTM E779 Compliant Software used for Multi-Point Test	
10	Corrected CFM50 (from software)	

D1. Altitude and Temperature Correction for Single-Point Test Data

01	Altitude and Temperature Correction Factor	
02	Corrected CFM50	

D2. Altitude and Temperature Correction for Multi-Point Test Data
Performed by blower door software.

E1. Accuracy Adjustment for Single-Point Test Data

01	Adjusted CFM50 (measured air leakage rate)	

E2. Accuracy Adjustment for Multi-Point Test Data

01	Percent Uncertainty @ 95% Confidence Level (from software)	
02	Accuracy Level	
03	Accuracy Adjustment Factor	
04	Adjusted CFM50 (measured air leakage rate)	

F. Compliance Statement

01	

G. Additional Requirements for Compliance

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The procedure for preparing the enclosure for testing is detailed in RESNET 380-2019 Section 4.2.
02	The procedure for installation of the test apparatus, and preparations for measurement shall conform to RESNET 380-2019 Section 4.3
03	The procedure for the conduct of the enclosure air leakage test shall conform to the One-Point Airtightness Test specified in RESNET 380-2019 Section 4.4.1.
04	The procedure for the conduct of the enclosure air leakage test shall conform to the Multi-Point Airtightness Test specified in RESNET 380-2019 Section 4.4.2.

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/AEA/ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this certificate of installation is true and correct.
2. I am either: a) a responsible person eligible under division 3 of the business and professions code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this certificate of installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person’s behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this certificate of installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the certificate of compliance, plans, and specifications approved by the enforcement agency.
4. I understand that an ECC-Rater will check the installation to verify compliance and if such checking determines the installation fails to comply, I am required to offer any necessary corrective action at no charge to the building owner.
5. I understand that a registered copy of this certificate of installation shall be posted or made available with the building permit(s) issued for the building and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this certificate of installation is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:
Third Party Quality Control Program (TPQCP) Status:	Name of TPQCP (if applicable):	

For assistance or questions regarding the Energy Standards, contact the Energy Hotline at: 1-800-772-3300

CF2R-ENV-20-H User Instructions

Section A. Enclosure Air Leakage – General Information

1. Select the appropriate test procedure. This selection will determine which sections of this document are required. Not that newer manometers have automatic functions for compensating baseline (automatic baseline) and compensating for house pressures other than the target. It is preferable to use these when available.
2. Enter the date that the enclosure air leakage test data was collected.
3. This field is automatically filled from the CF1R which determines if a CFM50 compliance target value is required.
4. This field determines the CFM50 target enclosure air leakage from the CF1R if ECC verification of enclosure air leakage is required.
5. Enter the indoor temperature measured at the time that the enclosure air leakage test was performed.
6. Enter the outdoor temperature measured at the time that the enclosure air leakage test was performed.
7. Provide a brief description of the location where the blower door was installed for the test. Examples: “front entry door on west side of house”, “door between house and garage”, “large window in family room”.
8. Enter the building elevation above sea level. Use the value for the closest city found in Reference Appendices, Joint Appendix JA2.2.

Section B. Diagnostic Equipment Information

1. Enter the number of manometers used to measure the enclosure pressurization. If more than one system is used, the fan flow numbers need to be manually added together, unless blower door software is used that will accommodate multiple fan systems running simultaneously.
2. Enter the make (brand) of the manometer used to collect the enclosure air leakage data. Examples: Retrotec, Energy Conservatory.
3. Enter the model of the manometer used to collect the enclosure air leakage data. Examples: DM-2 Mark II, DG700.
4. Enter the serial number of the manometer used to collect the enclosure air leakage data.
5. Enter the most recent date that the manometer was calibrated by following manufacturer’s calibration specifications.
6. This field is automatically filled. If the calibration date was more than 12 months prior to the test date entered in Row A02 above, an error will appear.
7. Enter the number of blower door fan systems required to run simultaneously to pressurize the enclosure for the enclosure air leakage test. If more than one system is used, the fan flow numbers need to be manually added together, unless blower door software is used that will accommodate multiple fan systems running simultaneously.
8. Enter the make (brand) of the fan used to collect the enclosure air leakage data. Examples: Retrotec, Energy Conservatory.
9. Enter the model of the fan used to collect the enclosure air leakage data. Examples: US1000, Q46, BD3, BD4.
10. Enter the serial number of the fan used to collect the enclosure air leakage data.
11. Enter the fan configuration shown on the meter. This is sometimes referred to as “range configuration”, “CONFIG” or “rings”. Examples: Open, A, B, C8.

Section C1. Enclosure Air Leakage Test for a Single-Point Test with Manual Meter

1. Enter the Time Average Period used on the manometer during the test. Must be at least 10 seconds.
2. Select the type of test being performed: Pressurization (air blowing into house) or depressurization (air blowing out of house).
3. Enter the pre-test baseline enclosure pressure. This is the reading on the manual manometer with no fans turned on.
4. This field is automatically calculated. This is the enclosure pressure target value the enclosure needs to achieve during the test.
5. Enter the unadjusted enclosure pressure measured. This value is read from the manual manometer during the test.
6. This field is automatically calculated. This value is the difference of the unadjusted enclosure pressure measured and the pre-test baseline enclosure pressure. The goal is to achieve 50 ± 3 Pa.
7. This field is automatically calculated. This field determines if the pressure achieved is acceptable to proceed with the enclosure air leakage test.
8. Enter the measured nominal fan flow at above fan pressure from the manometer that corresponds to the induced enclosure pressure difference.
9. This field is automatically calculated. The induced enclosure pressure difference is converted to a nominal airflow at 50 Pa.

Section C2. Enclosure Air Leakage Test for a Single-Point Test with Automatic Meter

1. Enter the time average period used on the manometer during the test. Must be at least 10 seconds.
2. Select the type of test being performed: Pressurization (air blowing into house) or depressurization (air blowing out of house).
3. Enter the pre-test baseline enclosure pressure. This is the reading on the automatic manometer with no fans turned on.
4. Enter the induced enclosure pressure from the automatic manometer. The goal is to achieve 50 ± 3 Pa.
5. This field is automatically calculated. This field determines if the pressure achieved is acceptable to proceed with the enclosure air leakage test.
6. Enter the measured nominal CFM50 from the automatic manometer.

Section C3. Enclosure Air Leakage Test for a Multi-Point Test

1. Enter the Time Average Period used on the manometer during the test. Must be at least 10 seconds.
2. Select the type of test being performed: Pressurization (air blowing into house) or depressurization (air blowing out of house).
3. Enter the pre-test baseline enclosure pressure. This is the reading on the manual manometer with no fans turned on.
4. This field is automatically calculated. This is the enclosure pressure target value the enclosure needs to achieve during the test.
5. Enter the unadjusted enclosure pressure measured. This value is read from the manual manometer during the test.
6. This field is automatically calculated. This value is the difference of the unadjusted enclosure pressure measured and the pre-test baseline enclosure pressure. The goal is to achieve 60 ± 3 Pa.
7. When using the software for a multi-point test, a minimum of five measures must be taken over a range of pressures. This is where the user acknowledges that this was done.
8. Enter the Post Test Baseline Enclosure Pressure from the manometer.

9. This version of the ENV-20 requires use of an ASTM E779-19 compliant software, typically provided by the blower door manufacturer. Confirm with the software vendor that it is compliant. Enter the name and version here.
10. Enter the final Corrected CFM50 reading from the software.

Section D1. Altitude and Temperature Correction for Single-Point Test Data

1. This field is automatically calculated. This factor is determined based on the altitude and temperature of the building location using equation 4 in Section 9 of ASTM E779-19.
2. This field is automatically calculated. The corrected CFM50 is the nominal CFM50 from Section C multiplied by the altitude and temperature correction factor.

Section D2. Altitude and Temperature Correction for Multi-Point Test Data

Section E1. Accuracy Adjustment for Single-Point Test Data

1. This field is automatically calculated. This value is determined from Equation 5a from ANSI/RESNET/ICC 380-2019.

Section E2. Accuracy Adjustment for Multi-Point Test Data

1. The software will provide a “Percent Uncertainty” value based on the readings taken. Enter that value here
2. This field is automatically calculated. If the Percent Uncertainty level is 10% or less, the Accuracy Level is “Standard”. If the Percent Uncertainty level is greater than 10%, the Accuracy Level is “Reduced”.
3. This field is automatically calculated:
    a. If the Accuracy Level is “Standard”, the Accuracy Adjustment Factor will be 1 (no adjustment)
    b. If the Accuracy Level is “Reduced”, the Accuracy Adjustment Factor will be adjusted by the Percent Uncertainty.
4. This field is automatically calculated. The Adjusted CFM50 is the Corrected CFM50 multiplied by the Accuracy Adjustment Factor.

Section F. Compliance Statement

1. This field is automatically calculated. A check is performed to make sure that the meter has been properly calibrated and that the measured enclosure air leakage is less than the target enclosure air leakage.

Section G. Additional Requirements for Compliance

1. This statement must be true (or not applicable) for the test to conform to the protocols.
2. This statement must be true (or not applicable) for the test to conform to the protocols.
3. This statement must be true (or not applicable) for the test to conform to the protocols.
4. This statement must be true (or not applicable) for the test to conform to the protocols.

A. Enclosure Air Leakage – General Information		
01	Test Procedure used	<< user pick text value from following list : **Single-Point Test with Manual Meter **Single-Point Test with Automatic Meter **Multi-Point Test>>
02	Date of the Diagnostic Test for this Dwelling	<<user input: date (use date format validation control)>>
03	Is ECC verification of building enclosure air leakage to outside required by CF1R?	<<calculated field: If CF1R requires ECC verification of building enclosure air leakage, then value = Required, else value = N/A>>
04	Target Enclosure Air Leakage from CF1R (CFM50)	<<if A03=required, then value = the CFM50 target value specified on the CF1R; else value = N/A>>
05	Indoor temperature during test (°F)	<<user input, numeric, x.x degF>>
06	Outdoor temperature during test (°F)	<<user input, numeric, x.x degF>>
07	Blower Door Location	<<user input, text, maximum 50 characters>>
08	Building Elevation Above Sea Level (ft)	<<user input, integer, xxxxx >>

B. Diagnostic Equipment Information

01	Number of Manometers Used to Measure Home Pressurization	<<user input, integer>> For entries >1, duplicate lines B. 2-6

02	03	04	05	06
Manometer Make	Manometer Model	Manometer Serial Number	Manometer Calibration Date	Manometer Calibration Status
<<user input, text, maximum 50 characters>>	<<user input, text, maximum 50 characters>>	<<user input, text, maximum 50 characters>>	<<user input, text (Date), maximum 50 characters>>	<<calculated field: if manometer Calibration Date in B05 is within 12 months of the date of the diagnostic test A02, then display message: Manometer Calibration is valid"; else display message: "WARNING - Manometer Calibration is expired. A manometer with current calibration is required in order to comply with this Enclosure Air Leakage Diagnostic test">>

07	Number of Fans Used to Pressurize Home	<<user input, integer>> For entries >1, duplicate lines B. 8-11

08	09	10	11
Fan Make	Fan Model	Fan Serial Number	Fan Configuration (rings)
<<user input, text, maximum 50 characters>>	<<user input, text, maximum 50 characters>>	<<user input, text, maximum 50 characters>>	<<user input, text, maximum 50 characters>>

C1. Enclosure Air Leakage Diagnostic Test for a Single-Point Test with Manual Meter
<< If A01 == Single-Point Test with Manual Meter this section is required>>

01	Time Average Period of Meter (seconds)	<<user enter integer ≥ 10>>
02	Test Methodology	<<user input, user pick one of the following 2 text values: **Pressurization; **Depressurization;>>
03	Pre-Test Baseline Enclosure Pressure (Pa) (May be positive or negative)	<<user enter numeric xx.x: -40 ≤value≥ 40>>
04	Unadjusted Enclosure Pressure Target (Pa)	<<calculated value: if C02 = Pressurization, then value =50 +C03; elseif C02=Depressurization then value=(-50) + C03>> (Resolution of 0.1)
05	Unadjusted Enclosure Pressure Measured (Pa) (Pressurization is positive; Depressurization is negative)	<<user enter numeric xx.x: -75.0 ≥ value ≤ 75>>
06	Induced Enclosure Pressure Difference (Pa)	<<calculated numeric value xx.x = C05-C03>>

	Goal = 50 ± 3 (Pressurization is positive; Depressurization is negative)	
07	Induced Enclosure Pressure Check	<<calculated value, if absolute value (C06) ≥ 15 Pa), display text: “Induced pressure within range for single point test; else display text: “Induced pressure too low for single point test - **Do Not Proceed**”>>
08	Measured Nominal Fan Flow at Above Fan Pressure (cfm) at the Induced Enclosure Pressure Difference (in C06 above)	<<user enter integer value, >> (Resolution of 1 CFM)
09	Calculated Nominal CFM50	<<calculated value, value = C08 * (50/[absolute value (C05 – C03)])$^{0.65}$ >>

C2. Enclosure Air Leakage Diagnostic Test for a Single-Point Test with Automatic Meter
<< If A01 == Single-Point Test with Automatic Meter this section is required>>

01	Time Average Period of Meter (seconds)	<<user enter integer ≥ 10>>
02	Test Methodology	<<user input, user pick one of the following 2 text values: **Pressurization; **Depressurization;>>
03	Pre-Test Baseline Enclosure Pressure (Pa) (May be positive or negative)	<<user enter numeric xx.x: -40 ≤value≥ 40>>
04	Induced Enclosure Pressure from Manometer (Pa) Goal = 50 ± 3 (Pressurization is positive; Depressurization is negative)	<<user enter numeric xx.x: -75.0 ≥ value ≤ 75>>
05	Induced Enclosure Pressure Check	<<calculated value, if absolute value (C04) ≥ 15 Pa, display text: “Induced pressure within range for single point test; else display text: “Induced pressure too low for single point test - **Do Not Proceed**”>>
06	Nominal CFM50	<<user enter integer>> (Resolution of 1 CFM)

C3. Enclosure Air Leakage Diagnostic Test for Multi-Point Test
<<If A01 == equals Multi-Point Test this section is required>>

01	Time Average Period of Meter (seconds)	<<user enter integer ≥ 10>>
02	Test Methodology	<<user input, user pick one of the following 2 text values: **Pressurization; **Depressurization;>>
03	Pre-Test Baseline Enclosure Pressure (Pa) (May be positive or negative)	<<user enter numeric xx.x: -40 ≤value≥ 40>>
04	Unadjusted Enclosure Pressure Target (Pa)	<<calculated value: if C02= Pressurization, then value =60 +C03; elseif C02=Depressurization then value=(-60) + C03>> (Resolution of 0.1)
05	Unadjusted Enclosure Pressure Measured (Pa) (Pressurization is positive; Depressurization is negative)	<<user enter numeric xx.x: -75.0 ≥ value ≤ 75>>
06	Induced Enclosure Pressure Difference (Pa) Goal = 60 ± 3 (Pressurization is positive; Depressurization is negative)	<<calculated numeric value xx.x = C05-C03>>
07	A minimum of five readings were taken spaced evenly between 10 Pa and 60 Pa (or highest attainable pressure)	<<user to acknowledge this requirement to continue>>
08	Post-Test Baseline Enclosure Pressure (Pa)	<<user enter value, >> (Resolution of 0.1 Pa)
09	Name and Version of ASTM E779 Compliant Software used for Multi-Point Test	<<user input, text, maximum 50 characters>>
10	Corrected CFM50 (from software)	<<user entry of value calculated by software >> (resolution of 1 CFM)

D1. Altitude and Temperature Correction for Single-Point Test Data
<<If A01 does not equal Multi-Point Test this section is required>>

a1	Outside Dynamic Viscosity of Air Correction **(this row is not visible to user)	<<if C02 = Depressurization, then value={2.629*10^{-3}*[(A06+460)$^{0.5}$]}/[1+(198.7/(A06+460))]; else report “NA”>>
a2	Indoor Dynamic Viscosity of Air Correction **(this row is not visible to user)	<< if C02 = Pressurization, then value={2.629*10^{-3}*[(A05+460)$^{0.5}$]}/[1+(198.7/(A05+460))]; else report “NA”>>
a3	Outdoor Air Density Correction **(this row is not visible to user)	<<if C02 = Depressurization, then value=0.07517*[1-((0.0035666*A08)/528)]$^{5.2553}$*(528/{A06+460}); else if C02 = Pressurization, then report “NA”>>
a4	Indoor Air Density Correction **(this row is not visible to user)	<<if C01 = Pressurization, then value=0.07517*[1-((0.0035666*A08)/528)]$^{5.2553}$*(528/{A05+460}); else report “NA”>>
01	Altitude and Temperature Correction Factor	<<if C02 = Depressurization, then value= [{(Da1)/0.044}$^{2*(0.65)-1}$] *[{(Da3)/0.07517}$^{1-0.65}$]; Else if CO2 = Pressurization, then value = [{(Da2)/0.044}$^{2*(0.65)-1}$] *[{(Da4)/0.07517}$^{1-0.65}$];
02	Corrected CFM50	<< If A01 equals Single-Point Test with Manual Meter value = C09 *D01 Else If A01 equals Single-Point Test with Automatic Meter value = C06 * D01>>

D2. Altitude and Temperature Correction for Multi-Point Test Data
Performed by blower door software.
<<If A01 equals Multi-Point Test this section is required>>

E1. Accuracy Adjustment for Single-Point Test Data
<<If A01 does not equal Multi-Point Test this section is required>>

01	Adjusted CFM50 (measured air leakage rate)	<<value = D02 * 1.1>>

E2. Accuracy Adjustment for Multi-Point Test Data
<<If A01 equals Multi-Point Test this section is required>>

01	Percent Uncertainty @ 95% Confidence Level (from software)	<<user entry of value calculated by software>>
02	Accuracy Level	<<calculated, if E01 ≤ 10, “Standard”; else if row E01 > 10, “Reduced”>>
03	Accuracy Adjustment Factor	<<calculated, if E02 = “Standard”, then value 1; else if E02 = “Reduced”, then value = 1 + (E01 % uncertainty/100)>>
04	Adjusted CFM50 (measured air leakage rate)	<<value = C10 * E03>>

F. Compliance Statement

01	<< if manometer Calibration Date in B05 is more than 12 months from the date of the diagnostic test A02 , then display text: “Fails Enclosure Air Leakage Test"; Else If A01 equals Single-Point Test with Manual Meter Or Single-Point Test with Automatic Meter If Adjusted CFM50 Leakage in E01 is less than or equal to the Building Air Leakage Rate Target in A04, then display text: " Passes Enclosure Air Leakage Test"; else display text: “Fails Enclosure Air Leakage Test”; Else If A01 equals Multi-Point Test If Adjusted CFM50 Leakage in E04 is less than or equal to the Building Air Leakage Rate Target in A04 then display text: "Passes Enclosure Air Leakage Test"; Else display text: " Fails Enclosure Air Leakage Test ">>

G. Additional Requirements for Compliance
<<Static Text. see the top>>

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met.